Comparison of Euler-Bernoulli and Timoshenko Beam Equations for Railway System Dynamics

A. Mosavi[1*], R. Benkreif [2], A. Varkonyi-Koczy [1,2]

[1] Obuda University, Institute of Automation, Kalman Kando Faculty of Electrical Engineering, 1034 Budapest, Becsi way 94-96., Hungary
[2] Department of Mathematics and Informatics, J. Selye University, Elektrarenska cesta 2, 945 01 Komarno, Slovakia

*amir.mosavi@kvk.uni-obuda.hu

Abstract. In railway system dynamics the dynamic stability problem has significant role particularly when it comes to dealing with the motion of the vertically deformable joints on damped Winkler foundation. Timoshenko and Euler-Bernoulli beam equations are the two widely used methods for dynamics analysis of this problem. This paper describes a comparison between Euler-Bernoulli and Timoshenko beam equations to investigate the track motion dynamic stability via solving the fourth order partial differential of the both models on an Elastic Foundation. This article aims at identifying an efficient model for future investigation on the track motion dynamics stability for the advanced railway systems.

Keywords: Railway System Dynamics, Euler-Bernoulli Beam, Timoshenko Beam

1 Introduction

There are numerous theories and modeling systems applied to the structural analysis and design of the beam for studying the railway system dynamics [1]. Among them the Euler-Bernoulli beam [2] is the most popular and widely used in the research community. Very recently, Wang (2015) [3], described the drawbacks with this theory in dealing with the deep beams. He further proposed the Timoshenko beam theory which is an advanced yet a simplified version of Euler–Bernoulli [4]. Today, the Timoshenko and Euler-Bernoulli beam model equations are the two widely used models for analyzing of the railway system dynamics. However in the novel model of Timoshenko the shear deformation forces have come into account, which were missing in Euler-Bernoulli model. In addition the cross section in Euler-Bernoulli theory stays stationary while in the Timoshenko's it rotates due to the shear forces [5]. Nevertheless in the both theories the reciprocal contact of the beam and the soil is generally the clue for solving the analysis modelling of the railway system dynamics. The spring elements are often the optimum replacement for the elastic foundation [6]. Further, the applied loads act

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in a transverse relation to the longitudinal axis as described in [7]. The deflection and internal forces of the beam are defined by the general theory of the beam [8].

1.1 Railway System Dynamics

The railway system dynamics includes two longitudinal parallel rails. They are fixed to the sleepers with a certain distance among them. The sleepers are fixed in the ground by a layer of ballast that is distributed all over the ground with a certain thickness [5]. The rail performance is about transferring the wheel load of the train to the sleepers while maintaining the rails in the suitable positions [6]. The load goes throw the sleeper from the rail and it is transmitted to the ballast [6]. Yet the transformation of the load depends on a lot of factors such as shape, size, and depth of ballast [9]. Fig.1 demonstrates the railway system dynamics, loads, and the stress distributions on the ballast. In Fig. 1, adopted from [10], $F_{zi\mathrm{R}}$, $F_{zi\mathrm{L}}$, $F_{yi\mathrm{L}}$, and $F_{yi\mathrm{R}}$ are the forces between the sleeper *i* and the left and right rails. In addition, the F_{ysbi} is the equivalent support force due to the ballast in the lateral direction. The d_r is the half distance between the left and right rails, d_b is the half distance between the centers of the left and right ballast bodies.

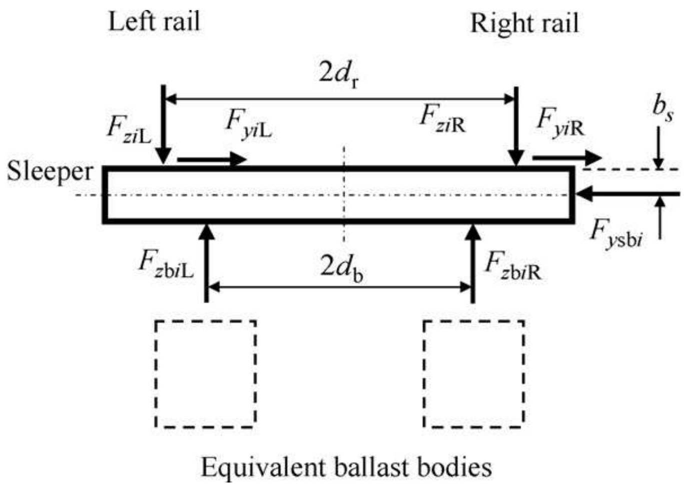

Fig. 1. Illustration of the railway system dynamics, loads, and the stress distributions on the ballast [10]

As Jin et al. (2008) states, by increasing the number of the wheels less bending moment is produced and the negative bending moment under neighboring wheel provides a relief of stress. According to Ling et al. (2014) the main factors effecting the value of the relief stress are the distance among the wheels and the range of the point of contraflexure of the rail. To design novel locomotives the railway system dynamics is to be carefully figured accordingly to well calculate the stress.

2 Timoshenko and Euler-Bernoulli beam equations

In solid mechanics there have been numerous theories introduced for structural modeling and analysis of beam [18,19]. Timoshenko beam [4,9] has been well studied and used for molding the railway system dynamics and analysis [20, 21,22]. Bogacz (2008) [23] describes that the main hypothesis for Timoshenko beam theory is that the unloaded beam of the longitudinal axis must be straight [8]. In addition the deformations and strains are considered to be small, and the stresses and strains can be modeled by Hook's law. Further to model the deformation [8], the plane of cross section will stay normal to the longitudinal axis and all applied loads act transversally. In 1921 the theory has been further advanced [10], then it was refined in 1922 by Timoshenko [4]. Since then it has been widely used for applications in vibration analysis of beam applied to railway system dynamics. Later, the nonlocal theory for vibration analysis of micro and Nano-structure [3, 23] has been the adjustment in Timoshenko theory. According to Kahrobaiyan (2014) there has been another effort to study beam vibration by utilizing the wave mechanics and it is not necessary if the beam material is isotropic and homogenous [9, 24]. The beam on elastic foundation model considered the spine of a lot of development over the years that made to track design [7,21,25]. In 1867 Winkler [8,13,26] introduced this model and it is still up until now in use for easy and quick deflection calculation of tracks. The mathematical formulation in the model is correct with a simple physical interpretation. Yet one can consider the rail modelled as an unlimited Euler-Bernoulli beam with an infinite longitudinal Winkler foundation [11], which considered as a boundless longitudinal support, separated by elastic vertical springs. The beam gets support by a distributed force that it is proportional with the beam deflection. The rail deflection z(x) could be obtained from the differential equation by only using two track parameters [11].

$$EI\,\frac{\partial^4 z(x)}{\partial x} + kz(x) = q(x) \tag{1}$$

Where, the real parameters z(x), x, q(x), EI and k are respectively, the rail deflection, the length coordinate, the distributed load on the rail, the beam bending stiffness EI (N/m^2) and the foundation stiffness ($\frac{N}{m^2}$,i.e. $\frac{N}{m}$ per Meter of rail).

3 Equation of Motion

Since 1921 the Timoshenko beam theory has been witnessed a great deal of development, and it has been used in modeling different scenarios [10]. Further, in 1948, the spread of waves in the transverse vibrations of bars and plates have determined by this theory. To obtain the solution for Timoshenko [10], the beam equation [8], and the Laplace transform was used. The operator of Timoshenko is the key to the system (1-4) that we aim at solving it.

$$\boldsymbol{a} \cdot \quad \boldsymbol{b} + \boldsymbol{c}^2\boldsymbol{\partial}_t^2 = (\boldsymbol{\partial}_x^2 - \boldsymbol{a}^2\boldsymbol{\partial}_t^2)(\boldsymbol{\partial}_x^2 - \boldsymbol{b}^2\boldsymbol{\partial}_t^2) + \boldsymbol{c}^2\boldsymbol{\partial}_t^2 \tag{2}$$

Fig. 2 illustrates the Winkler foundation using spring and damp for moving load in the railway road track showing the deformation of the beam [6]. The model is a moving loaded wheel on elastic foundation where a parallel connects linear Hertzian spring and damp the contact between the wheels and the rails [13]. This model is selected due to its relative simplicity, ease of calibration, and acceptance in engineering practice. As illustrated in Fig. 2, the soil-foundation interface is assumed to be an assembly of discrete, nonlinear elements composed of springs, dashpots and gap elements. It is intended to capture forces behavior of the ballast with a minimal number of parameters.

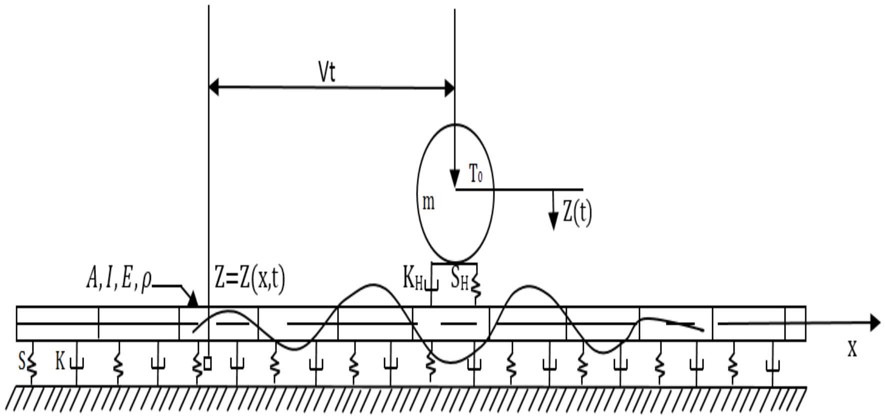

Fig. 2. Winkler foundation using spring and damp for moving load in the railway road track showing the deformation of the beam [6]

From Fig. the model shows that z (x, t) is the vertical displacement of the rail [12,3]. We want to get a solution for the model in the form:

$$z(x,t) = B(\xi)e^{wt} \tag{3}$$

where

$$\xi = x - yt, \text{and } e^{wt} = e^{(a+bi)t} = e^{at}(\cos(bt) + i.\sin(bt)) \tag{4}$$

Using the derivatives of z with respect of x and t, equation (5) simplifies to ODE and with substitution [14]. we get

$$EI\left[B^{IV} - (a^2+b^2)\left(w^2B\text{-}2\upsilon wB''' + \upsilon^2B^{IV}\right) + a^2b^2(w^2B\text{-}2\upsilon wB' + v^2B)\right] + \rho A(w^2B - 2\upsilon wB' + \upsilon^2B'') + k(wB - \upsilon B') + sB = \delta \tag{5}$$

We are looking for the characteristic polynomial of the Timoshenko model.

$$P_t(\lambda) = EI\lambda^4 \underbrace{+ \rho A(w - \upsilon\lambda)^2 + k(w - \upsilon\lambda) + s}_{P_{BE}(\lambda)} - (w - \lambda\upsilon)^2(\alpha\lambda^2 - \beta) \tag{6}$$

Where

$\alpha \coloneqq EI(a^2b^2)$ and $\beta \coloneqq EIa^2b^2$

The characteristic polynomial of E-B in the form:

$$P_{BE}(\mu) = \frac{4\rho AP_{BE}(\lambda)}{4\rho As - k^2} \tag{7}$$

We consider the roots of the polynomial above as functions hold with two parameters, namely v and w following the same method in [12]. As Zoller and Zobory (2002) [7] describe if we are looking for moderately damped, if and only if our relation:

$$4s(\rho A - \alpha + \beta) - k^2 \tag{8}$$

Using non-dimensional variables whish introduced in [3], we get

$$P(\mu) = \mu^4 + (c\mu - a - bi)^2(1 + \varepsilon_1 - \varepsilon_2\mu^2) + 1 \tag{9}$$

From the polynomial above we should use polynomial in the form

$$iP(i\mu) = 2ac\mu^2\varepsilon_2 - 2ab\varepsilon_2\mu^2 + 2ac\mu\varepsilon_1 - 2ab\varepsilon_1 + 2ac\mu - 2ab + i[-c^2\mu^4\varepsilon_2 + 2bc\mu^3\varepsilon_2 + a^2\mu^2\varepsilon_2 - b^2\mu^2\varepsilon_2 - c^2\mu^2\varepsilon_1 + 2c\mu b\varepsilon_1 - c^2\mu^2 + \mu^4 + a^2\varepsilon_1 - b^2\varepsilon_1 + 2c\mu b + a^2 - b^2 + 1 \tag{10}$$

By Routh-Hurwitz theorem [15].

$$f(z) = a_0z^n + b_0z^{n+1} + a_1z^{n-2} + b_1z^{n-3} + \cdots (a_0 \neq 0) \tag{11}$$

$$\left\|\begin{matrix} b_0 & b_1 & b_2 & \cdots & b_{n-1} \\ a_0 & a_1 & a_2 & \cdots & a_{n-1} \\ 0 & b_0 & b_1 & \cdots & b_{n-2} \\ 0 & a_0 & a_1 & \cdots & a_{n-2} \\ 0 & 0 & b_0 & \cdots & b_{n-3} \\ \cdots & \cdots & & \cdots & \\ \cdots & \cdots & & \cdots & \end{matrix}\right\| \quad \left(a_k = 0\ for\ k > \left[\frac{n}{2}\right], b_k = 0\ for\ k > \left[\frac{n-1}{2}\right]\right) \tag{12}$$

Using the matrix of Hurwitz [15], which is the matrix of square of order n and apply it in the polynomial (17).
The statement of theorem Det$A_8 \neq 0$ is proved.
The condition from the determinant of A = 0, a $\neq$0 and b = c = 0, then

$$\mu^2 = -\frac{1}{2}a^2\varepsilon_2 \pm i\frac{1}{2}\sqrt{a^4\varepsilon_2^2 - 4a^2\varepsilon_1 - 4a^2 - 4} \tag{13}$$

So, if our condition satisfied, then we have two roots in the left-hand and right-hand as well in the half plane.
Proof. If our polynomial (16) has no imaginary root, μ = ri and υ = bi, then we have new polynomial in the form

$$q(r) = r^4 + (cr - b)^2(r^2\varepsilon_2 + \varepsilon_1 + 1) + 1 \tag{14}$$

From our polynomial above we can get the possible values of b [14]. In our case the discriminant of b is 0, that imply the real roots (r) are in the range of the smooth real functions. The values of b it can write in the form [12].

$$b \pm (r) = \frac{cr^3\,\varepsilon_2 + cr\varepsilon_1 + cr \mp \sqrt{r^6\varepsilon_2 + r^4\varepsilon_1 + r^4 + r^2\varepsilon_2 + \varepsilon_1 + 1}}{r^2\varepsilon_2 + \varepsilon_1 + 1} \tag{15}$$

In the case if

$$c \geq \frac{\sqrt{2}\sqrt{(\varepsilon_1^2 + 2\varepsilon_1 + 1)(-\varepsilon - 2 + \sqrt{\varepsilon_1^2 + \varepsilon_2^2 + 2\varepsilon_1 + 1}}}{\varepsilon_1^2 + 2\varepsilon_1 + 1} \tag{16}$$

then, the functions b$\pm$ have zeroes and we always have real roots For the equation (15), the positive branch in the form

$$\lim_{n\to\infty} b_+(r) = +\infty \tag{17}$$

The negative branch in the form

$$\lim_{n\to\infty} b_{(-)}(r) = -\infty \tag{18}$$

Is satisfied.
Now if the derivative

$$b' \pm (r) = \left(3c\varepsilon_2 r^2 + c\varepsilon_1 + c + 1 + \frac{1}{2\frac{6r^5\varepsilon_2 + 4r^3\varepsilon_1 + 4r^3 + 2r\varepsilon_2}{\sqrt{r^6\varepsilon_2 + r^4\varepsilon_1 + r^4 + r^2\varepsilon_2 + \varepsilon_1 + 1}}}\right)(r^2\varepsilon_2 + \varepsilon_1 + 1)^{-1} - 2\frac{\left(c\varepsilon_2 r^2 + c\varepsilon_1 r + cr + \sqrt{r^6\varepsilon_2 + r^4\varepsilon_1 + r^4 + r^2\varepsilon_2 + \varepsilon_1 + 1}\right)r\varepsilon_2}{(r^2\varepsilon_2 + \varepsilon_1 + 1)^2} \tag{19}$$

is vanishing, then the extrema can be found. From this we obtain equation with respect to r. Using maple, we obtained solution for r. Min $b_+ = b_+(\sqrt{-r})$ and max $b_- = b_-(\sqrt{r})$. If c=0 and b $\geq$ 1, then the polynomial q can have imaginary roots, so b is in the range of the functions b$\pm$. Where the roots of μ = ri of p, two of them in the right-hand half plane and two of in the left-hand also, then q has two conjugate noimaginary root pairs. So, b is out of the range of functions b$\pm$.

Remark 1. The situation as to that of the un-damped classical case, so we can ignore the damping out if $Re(w) = \frac{-k}{2\rho A + \beta}$ where w is complex frequencies [13].

4 The solution of the system

Making the similar analysis adapted from [13], and implementing the method given by, Zobory and Zibolen [12]. Then, any non-dimensional speed c appears as critical speed [8], for some non-dimensional critical frequencies *n* [14].

If $$c = \frac{\sqrt{2}\sqrt{(\varepsilon_1^2+2\varepsilon_1+1)(-\varepsilon-2+\sqrt{\varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1+1}}}{\varepsilon_1^2+2\varepsilon_1+1} \quad (20)$$

Nor c = 0

Using the method introduced in [13]. From that we found the same result of the complex frequencies given by [16].

$$w_{1,2} = \frac{-k_H \pm i\sqrt{4ms_H - k_H^2}}{2m} \quad (21)$$

5 Conclusions

This article presents a comparison between Euler-Bernoulli and Timoshenko beam equations to investigate the track motion dynamic stability via solving the fourth order partial differential of the both models on an Elastic Foundation. It is described that the Timoshenko model, unlike the Euler-Bernoulli model, represents a deformable model. Further, it is concluded that the Timoshenko model has the supremacy in the beams with a low aspect ratio. Considering the restraints analysis of the models, the beam vibration of the Euler-Bernoulli produces the instability caused by the inadvertent excitation of un-modelled modes. In conclude, in Euler-Bernoulli beam theory the shear deformation is neglected and at longitudinal axis the plane section will stay normal. However, for Timoshenko model, the plane section will not be normal but it stays plane. Therefore, the shear deformation presents a variation among the plane section rotation and the normal to the longitudinal axis. As the result the Euler-Bernoulli beams theory is not able to consider the transverse shear stresses, while Timoshenko beams theory is capable of doing so. This research recognizes the Timoshenko model as an efficient way to investigate the track motion dynamics stability for the advanced railway system dynamics studies.

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